

Day 3 - Time Series and Sequential Decisions

Wednesday, July 9, 2025

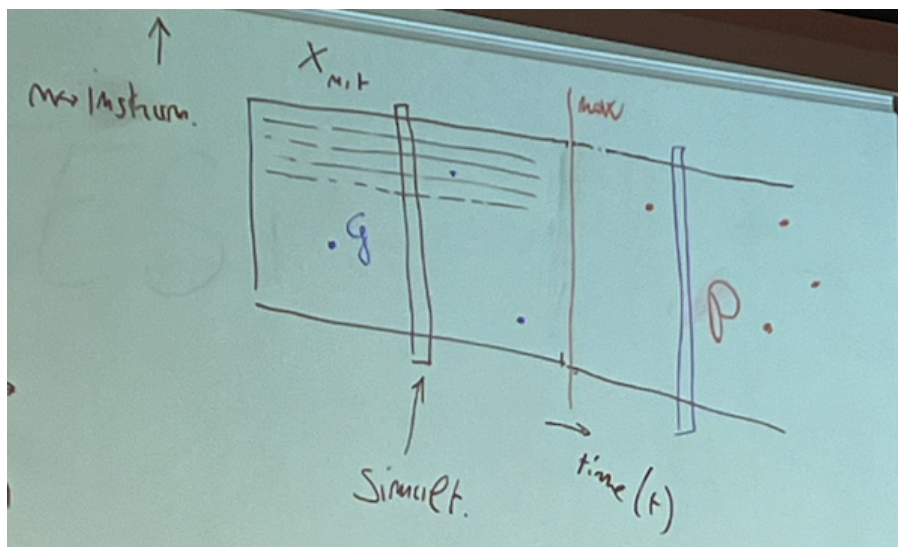
8:31 AM



Instead of X_n , the axis becomes X_t

Lab:

- 41.1, 41.2, 41.3
- 42.2 Random Walk
 - 42.2
 - 42.3
 - Video Arithmetic Brownian motion
 - Video Poisson process
 - 42.5
 - Video Stable levy process (cauchy increments)
- 43
 - 43.2
 - 43.3
 - Video 46.2
 - 43.4
- 44.1, 44.2, 44.4
- 45.3



- Each row is an instrument
- Each column is time that spans to the future infinitely
- Each column occurs simultaneously

Strong white noise	$\epsilon_t \sim \text{i.i.d.} \sim f(\epsilon).$ $\mathbb{L}\{\epsilon_t\} = 0$	$X_{t+1} = X_t + \epsilon_{t+1}$ $\quad = \epsilon_{t+1} + \epsilon_{t+2} + \dots$ $\mathbb{V}\{X_t\} = t\sigma$ $\text{stdev}\{X_t\} = \sqrt{t}\sigma$
Weak white noise	$\epsilon'_t \sim_{mc} \{0, \sigma_\epsilon^2\}$	

Covariance stationary theory

Wold $\mathbf{X}_t = \sum_{l=0}^{\infty} \beta_l \tilde{\epsilon}_{t-l}$	causal filter $\Leftrightarrow$	Cov-stationary process $\mathbf{X}_t$	Fourier tr. $\Leftrightarrow$	Cramer $\mathbf{X}_t = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{+\pi} e^{i\omega t} d\tilde{\mathbf{X}}_\omega$
$\Updownarrow$ Diagonal autocovariance	Gram-Schmidt $\Leftrightarrow$	$\Updownarrow$ Autocovariance	princ comp. $\Leftrightarrow$	$\Updownarrow$ Spectral density matrix
$\mathbb{I} \times \mathbf{1}_{\tau=0} = \mathbb{C}v\{\tilde{\epsilon}_t, \tilde{\epsilon}_{t+\tau}\}$		$\kappa_{\mathbf{X}}(\tau) = \mathbb{C}v\{\mathbf{X}_t, \mathbf{X}_{t+\tau}\}$		$\tilde{\kappa}_{\mathbf{X}}(\omega) d\omega = \mathbb{C}v\{d\tilde{\mathbf{X}}_\omega\}$

Glossary

Vocab	Explain
Covariance stationary process	(also called weakly stationary) has constant mean and covariance structure over time. Requirements: 1. Constant mean: $E[X_t] = \mu$ for all t 2. Finite variance: $\text{Var}(X_t) = \sigma^2 < \infty$ for all t 3. Covariance depends only on lag: $\text{Cov}(X_t, X_{t+k}) = \gamma(k)$ - depends only on gap k, not time t Key insight: The relationship between observations separated by k periods is the same regardless of when you measure it. vs Strictly stationary: Weaker condition - only requires first two moments to be time-invariant, not the entire distribution. Finance example: If stock returns are covariance stationary, the correlation between returns 5 days apart is the same whether you measure it in January or July.
Toeplitz pattern	a matrix where elements are constant along each diagonal. Structure: $\begin{bmatrix} a & b & c & d \\ e & a & b & c \\ f & e & a & b \\ g & f & e & a \end{bmatrix}$ Each diagonal (parallel to main diagonal) has identical elements. In covariance matrices: Covariance stationary processes have Toeplitz covariance matrices because $\text{Cov}(X_t, X_{t+k}) = \gamma(k)$ depends only on lag k. Example: If $\gamma(0)=1, \gamma(1)=0.5, \gamma(2)=0.2$, then: $\begin{bmatrix} 1.0 & 0.5 & 0.2 \\ 0.5 & 1.0 & 0.5 \\ 0.2 & 0.5 & 1.0 \end{bmatrix}$ Finance interpretation: Correlation structure repeats - relationship between observations k periods apart is always the same.
Cholesky	factors a positive definite matrix into the product of a lower triangular matrix and its

decomposition	transpose. Formula: $A = LL^T$ Where: • A: positive definite matrix (like covariance matrix) • L: lower triangular matrix • L^T: transpose of L (upper triangular) Key property: Unique decomposition for positive definite matrices. Finance applications: • Simulate correlated random variables: If $Z \sim N(0, I)$, then $LZ \sim N(0, A)$ • Portfolio risk decomposition • Monte Carlo simulation with specified correlation structure Example: Generate correlated stock returns with given covariance matrix by multiplying independent normal variables by Cholesky factor. Advantage: Computationally efficient and numerically stable.
Wold decomposition	states that any covariance stationary process can be written as the sum of two orthogonal components: $X_t = D_t + S_t$ Where: • D_t: deterministic component (predictable from past) • S_t: stochastic component (moving average of white noise) Stochastic part: $S_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ • Linear combination of current and past white noise shocks • ψ_j are MA coefficients with $\sum \psi_j^2 < \infty$ Key insight: Any stationary process can be represented as an infinite moving average ($MA(\infty)$) plus a predictable part. Finance application: Stock returns can be decomposed into predictable patterns and unpredictable shocks.
Convolution	combines two functions to produce a third function, showing how one function modifies another. Mathematical definition: $(f * g)(t) = \int f(\tau)g(t-\tau) d\tau$ Intuition: • Flip one function • Slide it across the other • Multiply and integrate at each position In time series: Convolution represents how past shocks (with different weights) combine to affect current values. Moving average example: $X_t = \sum \psi_j \varepsilon_{t-j}$ is a convolution of weights $\{\psi_j\}$ with white noise $\{\varepsilon_t\}$. Finance interpretation: Current asset price reflects weighted combination of all past news/shocks, with weights declining over time. Key property: Convolution in time domain = multiplication in frequency domain.